
PART A: EMPLOYABILITY SKILLS

UNIT 1: COMMUNICATION SKILLS-II

1. Communication and Its Importance

****Communication:**** The process of exchanging information, ideas, thoughts, and feelings between two or more people.

****Importance of Communication:****

- Helps in expressing thoughts and ideas.
- Builds relationships and trust.
- Improves teamwork and collaboration.
- Helps in decision-making and problem-solving.
- Essential for professional growth.

2. Types of Communication

****A. Verbal Communication:****

Communication using spoken words.

- ****Examples:**** Face-to-face conversations, phone calls, meetings, presentations.
- ****Advantages:**** Quick, immediate feedback, personal touch.
- ****Disadvantages:**** No record, can be misunderstood.

****B. Non-Verbal Communication:****

Communication without using words.

- ****Body Language:**** Posture, gestures, facial expressions.
- ****Eye Contact:**** Shows interest and confidence.
- ****Proximity:**** Personal space.
- ****Tone of Voice:**** Pitch, volume, speed.
- ****Advantages:**** Conveys emotions, reinforces verbal message.
- ****Disadvantages:**** Can be misinterpreted.

****C. Written Communication:****

Communication using written words.

- **Examples:** Emails, letters, reports, memos, messages.
- **Advantages:** Permanent record, clear, can be reviewed.
- **Disadvantages:** Time-consuming, no immediate feedback.

D. Visual Communication:

Communication using visual aids.

- **Examples:** Graphs, charts, maps, diagrams, images.
- **Advantages:** Easy to understand, engaging.
- **Disadvantages:** Can be complex to create.

3. Communication Styles

A. Passive Communication:

- Not expressing own needs or opinions.
- Avoids conflict, agrees with others.
- **Example:** "I don't mind, whatever you say."

B. Aggressive Communication:

- Expressing needs forcefully, ignoring others.
- Dominating, blaming, shouting.
- **Example:** "You are wrong! Do what I say."

C. Assertive Communication:

- Expressing needs clearly and respectfully.
- Listening to others, maintaining boundaries.
- **Example:** "I understand your point, but I think we should consider this."

D. Passive-Aggressive Communication:

- Indirectly expressing anger or frustration.
- Sarcasm, silent treatment, complaining.
- **Example:** "Fine, do whatever you want."

4. Barriers to Communication

A. Physical Barriers:

- Noise, distance, poor lighting, technical issues.

B. Language Barriers:

- Different languages, jargon, complex words.

****C. Psychological Barriers:****

- Stress, anger, fear, lack of confidence.

****D. Perceptual Barriers:****

- Misunderstandings, assumptions, biases.

****E. Cultural Barriers:****

- Different customs, beliefs, values.

****F. Organisational Barriers:****

- Hierarchy, policies, poor structure.

5. Effective Communication Skills

****A. Active Listening:****

- Paying full attention to the speaker.
- Not interrupting, asking questions.
- Summarising to confirm understanding.

****B. Clear and Concise:****

- Using simple, clear language.
- Avoiding jargon and complex words.

****C. Empathy:****

- Understanding others' feelings and perspectives.

****D. Confidence:****

- Speaking with assurance.
- Maintaining eye contact.

****E. Feedback:****

- Giving and receiving constructive feedback.

6. Writing Skills

****A. Importance of Writing:****

- Used in emails, reports, and documents.
- Creates a permanent record.

****B. Types of Writing:****

- ****Formal:**** Reports, business letters, official emails.
- ****Informal:**** Personal letters, casual messages.

****C. Tips for Good Writing:****

- Clear and concise.
- Correct grammar and spelling.
- Logical structure.
- Appropriate tone.

****D. Email Writing:****

- ****Subject line:**** Clear and specific.
- ****Greeting:**** Professional (Dear Sir/Madam).
- ****Body:**** Clear and concise.
- ****Closing:**** Thank you, Regards.
- ****Signature:**** Name, designation, contact.

UNIT 2: SELF-MANAGEMENT SKILLS-II

1. Self-Awareness

****Definition:**** Understanding your own emotions, strengths, weaknesses, values, and motivations.

****Importance:****

- Helps in making better decisions.
- Improves relationships.
- Increases self-confidence.
- Helps in identifying areas for improvement.

2. Self-Regulation

****Definition:**** Controlling your emotions, thoughts, and behaviour.

****Ways to Improve Self-Regulation:****

- Pause before reacting.
- Manage stress through deep breathing or exercise.
- Stay calm in difficult situations.

- Think before speaking.

3. Self-Motivation

****Definition:**** The ability to do what needs to be done without being influenced by others.

****Ways to Stay Motivated:****

- Set clear goals.
- Celebrate small achievements.
- Stay positive.
- Visualise success.
- Learn from failures.

4. Time Management

****Definition:**** Planning and controlling how much time to spend on specific activities.

****Importance:****

- Increases productivity.
- Reduces stress.
- Helps in achieving goals.
- Improves work-life balance.

****Techniques:****

- ****Prioritisation:**** Identify urgent and important tasks.
- ****To-Do Lists:**** Write down tasks for the day.
- ****Setting Deadlines:**** Complete tasks on time.
- ****Avoiding Procrastination:**** Start tasks early.
- ****Delegation:**** Assign tasks to others when possible.

5. Goal Setting

****Definition:**** Identifying what you want to achieve and creating a plan to achieve it.

****SMART Goals:****

- ****S**pecific:** Clear and well-defined.
- ****M**easurable:** Can track progress.
- ****A**chievable:** Realistic and attainable.

- **R**elevant: Aligned with your values and interests.
- **T**ime-bound: Has a deadline.

Example:

- I will complete my project report by Friday 5 PM.

6. Stress Management

Definition: Techniques used to cope with stress.

Causes of Stress:

- Workload, deadlines, personal issues.

Stress Management Techniques:

- **Exercise:** Physical activity reduces stress.
- **Breathing Exercises:** Deep breathing calms the mind.
- **Time Management:** Prevents last-minute pressure.
- **Healthy Lifestyle:** Proper diet and sleep.
- **Hobbies:** Activities that relax you.
- **Talking:** Discussing problems with friends or family.

UNIT 3: ICT SKILLS-II

1. Information and Communication Technology (ICT)

Definition: Technologies used to store, process, transmit, and retrieve information.

Components:

- **Hardware:** Computer, monitor, keyboard, printer.
- **Software:** Programs and applications.
- **Networks:** Internet, intranet, LAN, WAN.

2. Uses of ICT

In Communication:

- Email, video conferencing, social media, messaging apps.

****In Education:****

- Online learning, digital libraries, educational apps.

****In Business:****

- E-commerce, online banking, digital marketing.

****In Healthcare:****

- Electronic health records, telemedicine.

****In Entertainment:****

- Online gaming, streaming, music, movies.

3. Digital Devices

****A. Computers:****

- ****Desktop:**** Personal computer for office/home.
- ****Laptop:**** Portable computer.
- ****Tablet:**** Touchscreen device.
- ****Smartphone:**** Mobile phone with computing features.

****B. Input Devices:****

- Keyboard, mouse, scanner, microphone, touchscreen.

****C. Output Devices:****

- Monitor, printer, speakers, projector.

****D. Storage Devices:****

- Hard disk drive (HDD), solid-state drive (SSD), USB flash drive, memory card, cloud storage.

4. Internet and Its Uses

****Internet:**** A global network of interconnected computers.

****Uses of the Internet:****

- Communication (email, social media, chat).
- Information search (search engines like Google).
- Entertainment (videos, music, games).
- Education (online courses, e-books).
- E-commerce (online shopping, banking).

****WWW (World Wide Web):****

A collection of web pages on the internet.

****Browser:****

- A software used to access the internet.
- ****Examples:**** Chrome, Firefox, Edge, Safari.

****Search Engine:****

- A website that helps you search for information.
- ****Examples:**** Google, Bing, Yahoo, DuckDuckGo.

5. Digital Safety and Security

****A. Computer Security:****

- Protecting your computer from threats.
- ****Antivirus:**** Software that detects and removes viruses.
- ****Firewall:**** Prevents unauthorised access to your computer.
- ****Updates:**** Keep software updated to fix security issues.

****B. Password Security:****

- Use strong passwords (mix of letters, numbers, symbols).
- Do not share passwords.
- Change passwords regularly.
- Use two-factor authentication.

****C. Cyber Threats:****

- ****Virus:**** Malicious program that damages files.
- ****Malware:**** Software designed to harm computers.
- ****Phishing:**** Fake emails/messages to steal personal information.
- ****Hacking:**** Unauthorised access to computer systems.

****D. Safe Browsing:****

- Do not click on suspicious links.
- Use secure websites (look for "https").
- Avoid downloading from unknown sources.
- Use strong passwords and antivirus software.

****E. Social Media Safety:****

- Be careful about sharing personal information.
- Adjust privacy settings.
- Block/report inappropriate content.

UNIT 4: ENTREPRENEURIAL SKILLS-II

1. Entrepreneurship

****Definition:**** The process of starting and running a business to make a profit.

****Entrepreneur:**** A person who starts a new business, taking financial and personal risks.

****Importance:****

- Creates jobs.
- Drives innovation.
- Contributes to economic growth.
- Solves problems and improves lives.

2. Types of Business

****A. Micro Business:****

- Very small business (e.g., street vendor, home-based business).
- Minimal investment and employees.

****B. Small Business:****

- Small-scale business with limited employees.
- ****Examples:**** Local shops, restaurants, service providers.

****C. Large Business:****

- Large-scale business with many employees.
- ****Examples:**** MNCs, corporations.

3. Business Opportunities

****Identifying Business Opportunities:****

- Identify problems that need solving.
- Look for gaps in the market.
- Consider your skills and interests.
- Research trends and demands.

****Examples:****

- Online tutoring, food delivery, e-commerce, digital marketing.

4. Business Planning

****A. Business Plan:****

A written document that describes the business idea, goals, strategies, and financial projections.

****Components:****

- ****Executive Summary:**** Overview of the business.
- ****Business Description:**** What the business does.
- ****Market Analysis:**** Target customers, competitors.
- ****Organisational Structure:**** Roles and responsibilities.
- ****Product/Service:**** What is offered.
- ****Marketing Strategy:**** How to attract customers.
- ****Financial Plan:**** Budget, projections, funding.
- ****Risk Analysis:**** Potential challenges.

5. Financial Literacy

****A. Revenue and Expenses:****

- ****Revenue:**** Money earned from selling products/services.
- ****Expenses:**** Money spent on running the business.

****B. Profit and Loss:****

- ****Profit:**** Revenue - Expenses (positive).
- ****Loss:**** Revenue - Expenses (negative).

****C. Cash Flow:****

Movement of money in and out of the business.

****D. Funding Sources:****

- Personal savings.
- Loans from banks.
- Investors.
- Government schemes.

6. Risk Management

****Definition:**** Identifying and managing potential risks.

****Types of Risks:****

- Financial risk (loss of money).
- Operational risk (issues with operations).
- Market risk (changes in demand).
- Legal risk (lawsuits).

****Risk Management Strategies:****

- Avoid the risk.
- Reduce the risk.
- Transfer the risk (insurance).
- Accept the risk.

UNIT 5: GREEN SKILLS-II

1. Sustainable Development

****Definition:**** Development that meets the needs of the present without compromising the ability of future generations to meet their own needs.

****Importance:****

- Conserves natural resources.
- Protects the environment.
- Ensures future generations have a better quality of life.

2. Green Economy

****Definition:**** An economy that is low-carbon, resource-efficient, and socially inclusive.

****Characteristics:****

- Uses renewable energy.
- Reduces pollution.
- Promotes energy efficiency.
- Creates green jobs.

****Green Jobs:****

Jobs that contribute to preserving or restoring the environment.

- ****Examples:**** Solar panel installer, wind turbine technician, environmental engineer, waste management specialist.

3. Natural Resources

****Types:****

****A. Renewable Resources:****

- Can be replenished naturally.
- ****Examples:**** Solar energy, wind energy, water, forests.

****B. Non-Renewable Resources:****

- Cannot be replenished easily.
- ****Examples:**** Coal, oil, natural gas, minerals.

4. Environmental Issues

****A. Climate Change:****

- Changes in global temperatures and weather patterns.
- ****Causes:**** Greenhouse gas emissions (CO₂, methane).
- ****Effects:**** Rising sea levels, extreme weather, loss of biodiversity.

****B. Pollution:****

- ****Air Pollution:**** Vehicle emissions, industrial smoke.
- ****Water Pollution:**** Industrial waste, sewage, plastics.
- ****Soil Pollution:**** Chemicals, pesticides, waste.

****C. Deforestation:****

- Clearing of forests for agriculture, industry.
- ****Effects:**** Loss of biodiversity, climate change.

****D. Waste Management:****

- ****Biodegradable Waste:**** Decomposes naturally (e.g., food, paper).
- ****Non-Biodegradable Waste:**** Does not decompose (e.g., plastics, glass).
- ****3Rs:**** Reduce, Reuse, Recycle.

5. Sustainable Practices

A. Energy Conservation:

- Use energy-efficient appliances.
- Turn off lights and electronics when not in use.
- Use renewable energy sources.

B. Water Conservation:

- Avoid wasting water.
- Use rainwater harvesting.
- Treat wastewater.

C. Waste Management:

- Composting organic waste.
- Recycling paper, plastic, glass.
- Segregating waste (wet/dry).

D. Tree Plantation:

- Plant more trees.
- Protect existing forests.

E. Eco-Friendly Products:

- Use biodegradable products.
- Avoid single-use plastics.

PART B: SUBJECT SPECIFIC SKILLS

UNIT 1: DIGITAL DOCUMENTATION (ADVANCED) USING LIBREOFFICE WRITER

CHAPTER 1: INTRODUCTION TO STYLES

1. What are Styles?

****Styles:**** A set of formatting attributes (font, size, colour, alignment, etc.) that can be applied to text, paragraphs, pages, and other elements in a document.

****Advantages of Using Styles:****

- ****Consistency:**** Uniform formatting throughout the document.
- ****Efficiency:**** Apply formatting quickly.
- ****Easy Updates:**** Change the style once, and it updates everywhere.
- ****Professional Appearance:**** Creates well-structured documents.

2. Style Categories in Writer

****A. Page Style:****

- Controls the formatting of pages (margins, headers, footers, page numbers, borders).
- ****Examples:**** First Page, Default Page Style, Left Page, Right Page.

****B. Paragraph Style:****

- Controls the formatting of paragraphs (alignment, indentation, spacing, font, size, line spacing, borders).
- ****Examples:**** Default Style, Heading 1, Heading 2, Body Text, Quotations.

****C. Character Style:****

- Controls the formatting of selected characters or words within a paragraph (font, size, colour, bold, italic, underline).
- ****Examples:**** Emphasis, Strong Emphasis, Teletype, Superscript, Subscript.

****D. Frame Style:****

- Controls the formatting of frames (text boxes, images, objects) – borders, background, position, wrapping.
- ****Examples:**** Frame, Graphics, OLE (Object Linking and Embedding).

****E. List Style:****

- Controls the formatting of bulleted and numbered lists (bullets, numbering, indentation, spacing).
- ****Examples:**** Numbering 1, Numbering 2, Bullet, Bullet 2.

****F. Table Style:****

- Controls the formatting of tables (borders, background, fonts, alignment, grid lines).
- ****Examples:**** Default Table Style, Elegant, Simple.

3. Styles and Formatting Window

****How to Open:****

- Press F11.

- Or go to Format → Styles and Formatting.
- Or click the Styles and Formatting icon in the toolbar.

****Components:****

- ****Style Categories:**** Buttons at the top to switch between styles.
- ****List of Styles:**** Displays available styles in the selected category.
- ****Fill Format Mode:**** Apply the same style to multiple areas.
- ****New Style from Selection:**** Create a new style from selected text.
- ****Update Style:**** Update an existing style with new attributes.
- ****Load Styles:**** Import styles from another document or template.

4. Creating a New Style

****Method 1: From Selection****

1. Format the text (font, size, colour, alignment, etc.).
2. Select the formatted text.
3. In the Styles and Formatting window, click the "New Style from Selection" icon.
4. Enter a name for the new style.
5. Click OK.

****Method 2: Drag and Drop****

1. Select the formatted text.
2. Drag the selection and drop it into the Styles and Formatting window.
3. Enter a name for the new style.
4. Click OK.

5. Updating a Style

****Method 1: Update from Selection****

1. Select the text with the modified formatting.
2. In the Styles and Formatting window, right-click on the style you want to update.
3. Select "Update Style."

****Method 2: Modify Style****

1. In the Styles and Formatting window, right-click on the style.
2. Select "Modify."
3. Make the desired changes in the dialog box.
4. Click OK.

6. Loading Styles from Template or Document

1. Go to Format → Styles and Formatting.
2. Click the "Load Styles" icon.
3. In the Load Styles dialog box:
 - Click "From File" to browse for a template or document.
 - Select the styles you want to load.
 - Check "Overwrite" if you want to replace existing styles with the same name.
4. Click OK.

7. Applying Styles

1. Select the text or paragraph.
2. In the Styles and Formatting window, double-click on the desired style.
3. Or, from the toolbar drop-down, select the style.

8. Fill Format Mode

1. In the Styles and Formatting window, click the "Fill Format Mode" icon.
2. The cursor changes to a paintbrush.
3. Select the style you want to apply from the list.
4. Click on the areas of the document where you want to apply the style.
5. To exit Fill Format Mode, click the "Fill Format Mode" icon again.

CHAPTER 2: WORKING WITH IMAGES

1. Inserting an Image

****Methods:****

****A. Insert Image Option:****

1. Place the cursor where you want to insert the image.
2. Go to Insert → Image.
3. Browse and select the image file.
4. Click Open.

****B. Drag and Drop:****

1. Open the folder containing the image.
2. Drag the image file and drop it into the document.

****C. Copy and Paste:****

1. Right-click on the image file and select Copy.
2. In the document, right-click and select Paste.

****D. Inserting an Image by Linking:****

1. Go to Insert → Image.
 2. In the dialog box, check "Link."
 3. Browse and select the image file.
 4. Click Open.
- ****Advantage:**** The document size remains small.
 - ****Disadvantage:**** The image must remain in the same location.

2. Modifying an Image

****A. Image Toolbar:****

- The Image toolbar appears when an image is selected.
- ****Tools:**** Resize, crop, rotate, flip, change colour, adjust transparency.

****B. Resizing an Image:****

- Select the image, drag the corner handles to resize.
- Keep aspect ratio: Hold the Shift key while dragging.

****C. Cropping an Image:****

1. Select the image.
2. In the Image toolbar, click the "Crop" icon.
3. Drag the handles to select the area to keep.
4. Press Enter to crop.

****D. Deleting an Image:****

- Select the image and press the Delete key.

3. Drawing Objects

****A. Creating Drawing Objects:****

1. Go to View → Toolbars → Drawing.

2. Click the desired shape (line, rectangle, ellipse, text box, etc.).
3. Draw the shape on the document.

****B. Setting or Changing Properties:****

- Select the object.
- Right-click → Object Properties or use the Drawing Object Properties toolbar.
- Change: Fill colour, line colour, line style, transparency.

****C. Resizing and Grouping:****

- ****Resize:**** Select the object, drag the handles.
- ****Grouping:****
 1. Select multiple objects (Shift + Click).
 2. Right-click → Group → Group.

4. Positioning Images in Text

****A. Arrangement:****

- ****Bring to Front:**** Places the image above other objects.
- ****Send to Back:**** Places the image behind other objects.
- ****Bring Forward/Send Backward:**** Moves the image one step forward/backward.

****B. Anchoring:****

- ****To Page:**** The image stays at a fixed position on the page.
- ****To Paragraph:**** The image moves with the paragraph.
- ****To Character:**** The image is treated as a character.
- ****As Character:**** The image behaves like a character in the text.

****C. Alignment:****

- ****Align Left, Center, Right:**** Horizontal alignment.
- ****Align Top, Bottom, Center:**** Vertical alignment.

****D. Text Wrapping:****

- ****None:**** Text does not wrap around the image.
- ****Page Wrap:**** Text flows around the image.
- ****Optimal Page Wrap:**** Text flows around the image but avoids gaps.
- ****Wrap Through:**** Text goes through the image (like a watermark).
- ****Wrap Left/Right:**** Text wraps only on one side.
- ****In Background:**** Image is in the background.

CHAPTER 3: ADVANCED FEATURES OF WRITER

1. Table of Contents (ToC)

****Definition:**** A list of headings and their page numbers at the beginning of a document.

****Creating a ToC:****

1. Apply heading styles (Heading 1, Heading 2, Heading 3) to the headings in the document.
2. Place the cursor where you want the ToC.
3. Go to Insert → Table of Contents and Index → Table of Contents, Index or Bibliography.
4. Select the type (Table of Contents).
5. Click OK.

****Customising a ToC:****

1. Right-click on the ToC.
2. Select "Edit Index/Table."
3. In the dialog box, you can:
 - Change the title.
 - Select the heading levels to include.
 - Change the format.
 - Add tab stops and page numbers.

****Updating a ToC:****

1. Right-click on the ToC.
2. Select "Update Index/Table."

****Deleting a ToC:****

1. Right-click on the ToC.
2. Select "Delete Index/Table."

2. Templates

****Definition:**** A pre-designed document that contains styles, formatting, and content.

****Creating a Template:****

1. Create a document with the desired styles and content.
2. Go to File → Templates → Save as Template.
3. Enter a name for the template.
4. Click OK.

****Using a Template:****

1. Go to File → New → Templates.
2. Select the template.
3. Click Open.

****Editing a Template:****

1. Go to File → Templates → Edit.
2. Select the template.
3. Make the changes.
4. Save the template.

3. Track Changes

****Definition:**** A feature that records all edits made to a document, allowing you to review and accept or reject them.

****Turning on Track Changes:****

1. Go to Edit → Track Changes → Record.
2. Or press Ctrl + Shift + E.

****Viewing Changes:****

- Deleted text: Shown in colour with strikethrough.
- Added text: Shown in colour with underline.
- Formatting changes: Shown in colour with margin notes.

****Accepting/Rejecting Changes:****

1. Go to Edit → Track Changes → Manage.
2. Select a change.
3. Click Accept or Reject.
4. Or, right-click on the change and select Accept or Reject.

4. Comments

****Adding a Comment:****

1. Select the text.
2. Go to Insert → Comment.
3. Or press Ctrl + Alt + C.
4. Type the comment.

****Editing a Comment:****

1. Double-click on the comment.

2. Edit the text.

****Deleting a Comment:****

1. Right-click on the comment.
2. Select Delete Comment.

5. Comparing Documents

****Comparing Two Documents:****

1. Go to Edit → Track Changes → Compare Documents.
2. Select the two documents to compare.
3. The differences will be shown as tracked changes.

UNIT 2: ELECTRONIC SPREADSHEET (ADVANCED) USING LIBREOFFICE CALC

CHAPTER 4: ANALYSE DATA USING SCENARIOS AND GOAL SEEK

1. Consolidating Data

****Definition:**** Combining data from multiple sheets or ranges into a single sheet.

****Steps:****

1. Go to Data → Consolidate.
2. Select the function (Sum, Average, Count, etc.).
3. Add the ranges to be consolidated.
4. Choose options (Use row labels, Use column labels).
5. Click OK.

2. Groups and Subtotals

****A. Creating Groups:****

1. Select the rows or columns to group.
2. Go to Data → Group and Outline → Group.
3. Or press F12.

****B. Adding Subtotals:****

1. Sort the data by the column you want to group by.
2. Go to Data → Subtotals.
3. Select the Group By column.
4. Select the function (Sum, Average, Count).
5. Select the column to calculate.
6. Click OK.

3. What-If Analysis

****Definition:**** The process of changing the values in cells to see how those changes affect the outcome of formulas.

4. Scenarios

****Definition:**** A set of input values that you can save and use to analyse different possibilities.

****Creating a Scenario:****

1. Go to Tools → Scenarios.
2. Click New.
3. Enter a name for the scenario.
4. Select the cells that will change.
5. Enter the values.
6. Click OK.

****Using a Scenario:****

1. Go to Tools → Scenarios.
2. Select the scenario.
3. Click Apply.

5. Goal Seek

****Definition:**** A tool that finds the input value needed to achieve a desired output.

****Steps:****

1. Go to Tools → Goal Seek.
2. In the "Formula cell" field, enter the cell that contains the formula.

3. In the "Target value" field, enter the desired result.
4. In the "Variable cell" field, enter the cell that you want to change.
5. Click OK.

****Example:****

If you want to find the interest rate needed to make the monthly payment Rs. 5000 on a loan.

- Formula cell: Payment cell
- Target value: 5000
- Variable cell: Interest rate cell

CHAPTER 5: USING MACROS

1. What is a Macro?

****Definition:**** A recorded sequence of actions that can be played back to automate repetitive tasks.

****Advantages:****

- Saves time.
- Reduces errors.
- Automates routine tasks.

2. Recording a Macro

1. Go to Tools → Macros → Record Macro.
2. Perform the actions you want to record.
3. Click the Stop Recording button.
4. Enter a name for the macro.
5. Choose where to save it (My Macros or the document).
6. Click Save.

3. Running a Macro

1. Go to Tools → Macros → Run Macro.
2. Select the macro.
3. Click Run.

4. Creating a Simple Macro

****Example:****

1. Record a macro that formats a selected cell to Bold and Red.
2. Save it as "FormatBoldRed."

5. Macro as a Function

****Creating a Custom Function:****

1. Go to Tools → Macros → Organise Macros → OpenOffice Basic.
2. Write the function code in the Basic editor.
3. Save and close.

****Example:****

```
Function AddTwoNumbers(a, b)
    AddTwoNumbers = a + b
End Function
```

****Using the Function:****

- In a cell: =AddTwoNumbers(5, 3) → Result: 8

CHAPTER 6: LINKING SPREADSHEET DATA

1. Setting Up Multiple Sheets

****Adding a New Sheet:****

- Click the "+" button at the bottom left of the window.
- Or go to Insert → Sheet.

****Renaming a Sheet:****

- Double-click on the sheet tab.
- Type the new name.

2. Creating References to Other Sheets

****Using Keyboard and Mouse:****

1. In the formula bar, type "=".
2. Click on the sheet tab of the other sheet.
3. Click on the cell you want to reference.
4. Press Enter.

****Syntax:****

- =Sheet2.A1

3. Creating References to Another Document

1. Open both documents.
2. In the formula bar, type "=".
3. Switch to the other document window.
4. Click on the cell you want to reference.
5. Press Enter.

****Syntax:****

- ='[DocumentName.ods]Sheet1'!A1

4. Hyperlinks

****Creating a Hyperlink:****

1. Select the cell or text.
2. Go to Insert → Hyperlink.
3. Select the type:
 - ****Internet:**** Website URL.
 - ****Mail & News:**** Email address.
 - ****Document:**** Link to another cell or sheet in the same document.
 - ****New Document:**** Create a new document.
4. Enter the target and text.
5. Click Apply.

****Editing a Hyperlink:****

1. Right-click on the hyperlink.
2. Select Edit Hyperlink.

****Deleting a Hyperlink:****

1. Right-click on the hyperlink.
2. Select Remove Hyperlink.

5. Linking to External Data

1. Go to Insert → Sheet from File.
2. Select the file.
3. Choose the sheets to import.
4. Choose the options (Link to external data).
5. Click OK.

6. Linking to Registered Data Sources

1. Go to View → Data Sources (F4).
2. Click the "Register Data Sources" icon.
3. Select the data source type (e.g., Database, Spreadsheet).
4. Browse and select the file.
5. Click OK.

CHAPTER 7: SHARE AND REVIEW A SPREADSHEET

1. Sharing a Spreadsheet

****Sharing Options:****

- ****Save in a Shared Folder:**** Others can access the file.
- ****Send via Email:**** Attach the file.
- ****Cloud Storage:**** Google Drive, OneDrive, Dropbox.

2. Opening and Saving a Shared Spreadsheet

- ****Open:**** Use File → Open or double-click.
- ****Save:**** File → Save or Ctrl + S.

3. Recording Changes

****Turning on Record Changes:****

1. Go to Edit → Track Changes → Record.
2. Or press Ctrl + Shift + E.

****Viewing Changes:****

- Changes are highlighted and shown in the margin.

4. Adding, Editing, and Formatting Comments

****Adding a Comment:****

1. Select the cell.
2. Go to Insert → Comment.
3. Or right-click → Insert Comment.
4. Type the comment.

****Editing a Comment:****

1. Double-click on the comment.
2. Edit the text.

****Formatting a Comment:****

1. Right-click on the comment.
2. Select Format Comment.
3. Change the background, font, size, etc.

5. Reviewing Changes

****Accepting/Rejecting Changes:****

1. Go to Edit → Track Changes → Manage.
2. Select a change.
3. Click Accept or Reject.

6. Merging and Comparing Sheets

1. Go to Edit → Track Changes → Merge Documents.
2. Select the document to merge.

3. The changes are merged, and conflicts are highlighted.

UNIT 3: DATABASE MANAGEMENT SYSTEM USING LIBREOFFICE BASE

CHAPTER 8: INTRODUCTION TO DATABASE MANAGEMENT SYSTEM

1. Data and Information

****Data:****

- Raw facts, figures, symbols, or events.
- ****Example:**** 25, "John", 5/10/2024

****Information:****

- Processed data that is meaningful and useful.
- ****Example:**** John is 25 years old and joined on 5/10/2024.

2. Database and DBMS

****Database:****

- An organised collection of related data.
- ****Example:**** Employee database, student database, library database.

****DBMS (Database Management System):****

- Software that manages databases.
- ****Functions:**** Storing, retrieving, updating, and managing data.
- ****Examples:**** MySQL, Oracle, SQL Server, LibreOffice Base.

3. Advantages of Database

- ****Reduced Redundancy:**** Data is stored only once.
- ****Data Consistency:**** Accurate and up-to-date data.
- ****Data Security:**** Access control and permissions.
- ****Data Integrity:**** Enforces rules and constraints.
- ****Data Sharing:**** Multiple users can access data.

- **Backup and Recovery:** Protects data from loss.

4. Data Models

A. Hierarchical Data Model:

- Data is organised in a tree-like structure (parent-child).
- **Example:** File system.

B. Network Data Model:

- Data is organised in a graph structure (many-to-many relationships).
- **Example:** Telecommunications network.

C. Relational Data Model:

- Data is organised in tables (relations).
- Tables are connected using keys.
- **Example:** Most modern databases (MySQL, Oracle, Base).

5. Relational Database Model

Key Concepts:

A. Table (Relation):

- A structured collection of data with rows and columns.
- **Example:** Employee table.

B. Field (Column/Attribute):

- A single piece of information in a table.
- **Example:** Emp_ID, Name, Department.

C. Record (Row/Tuple):

- A complete set of fields.
- **Example:** A single employee's details.

D. Key Types:

Primary Key:

- A unique identifier for each record.
- Cannot be NULL.
- **Example:** Emp_ID.

****Foreign Key:****

- A field that refers to the primary key of another table.
- Creates relationships between tables.
- ****Example:**** Dept_ID in Employee table references Dept_ID in Department table.

****Composite Key:****

- A combination of two or more fields that uniquely identifies a record.

****Candidate Key:****

- A field or combination of fields that can serve as a primary key.

6. Objects of RDBMS

****A. Tables:****

- Store data.

****B. Queries:****

- Ask questions (retrieve data based on conditions).

****C. Forms:****

- User-friendly interfaces for data entry.

****D. Reports:****

- Formatted output of data (printable).

****E. Macros:****

- Automate tasks (recorded actions).

CHAPTER 9: STARTING WITH LIBREOFFICE BASE

1. Introduction to LibreOffice Base

****What is LibreOffice Base?****

- A free and open-source database management system.
- It allows you to create and manage databases, create tables, queries, forms, and reports.

2. Data Types

****Definition:**** The kind of data a field can hold.

****Common Data Types:****

- ****Text:**** Alphanumeric characters (name, address).
- ****Integer (INT):**** Whole numbers (age, quantity).
- ****Decimal (NUMERIC/DECIMAL):**** Numbers with decimals (price, salary).
- ****Date:**** Date values (date of birth, joining date).
- ****Time:**** Time values.
- ****Boolean (YES/NO):**** True or false values.
- ****Varchar:**** Variable-length text.
- ****Long Varchar:**** Longer text (large text fields).

3. User Interface of LibreOffice Base

****Components:****

- ****Database Pane:**** Lists tables, queries, forms, reports.
- ****Tasks Pane:**** Lists tasks for creating objects.
- ****Menu Bar:**** File, Edit, View, Insert, Tools, etc.
- ****Toolbar:**** Quick access to common actions.

4. Creating a Table

****Method 1: Using Wizard****

1. In the Database Pane, select Tables.
2. In the Tasks Pane, click "Use Wizard to Create Table."
3. Select fields from the available categories.
4. Add fields to the selected fields list.
5. Click Next.
6. Assign field names and data types.
7. Set the primary key.
8. Click Finish.

****Method 2: Using Design View****

1. In the Database Pane, select Tables.
2. In the Tasks Pane, click "Create Table in Design View."
3. Enter:
 - ****Field Name:**** Name of the field.
 - ****Field Type:**** Select the data type.

- **Description:** (Optional).
- 4. Set the primary key:
 - Right-click on the row and select "Primary Key."
- 5. Save the table (Ctrl + S).

5. Setting Primary Key

Using Wizard:

- In the wizard, select "Create a primary key" and choose the field.

Using Design View:

- Right-click on the field and select "Primary Key."

Note: The primary key field should contain unique values (not NULL).

6. Saving a Table

- Click File → Save.
- Or press Ctrl + S.
- Enter the table name.
- Click OK.

7. Entering Data in the Table

1. Open the table (double-click in the Database Pane).
2. Click in the first empty row.
3. Enter data in each field.
4. Press Enter to move to the next row.

8. Navigating through the Table

- Use the arrow keys.
- Use the Navigation Bar at the bottom:
 - First Record, Previous Record, Next Record, Last Record.
 - New Record, Delete Record.

9. Editing Data

1. Click on the cell with the data.
2. Edit the content.
3. Press Enter to save.

10. Deleting Records

1. Click on the record selector (leftmost column).
2. Right-click → Delete.
3. Confirm deletion.

11. Sorting Data

1. Click on the column header.
2. Click the Sort Ascending or Sort Descending button.
3. Or go to Tools → Sort.

CHAPTER 10: WORKING WITH MULTIPLE TABLES

1. Editing and Deleting Tables

Editing a Table:

1. Right-click on the table in the Database Pane.
2. Select Edit.
3. Make changes (add/delete fields, change data types).
4. Save the table.

Deleting a Table:

1. Right-click on the table in the Database Pane.
2. Select Delete.

2. Relationships between Tables

****Definition:**** Connections between tables based on a common field.

****Types of Relationships:****

****A. One-to-One:****

- Each record in Table A is related to exactly one record in Table B.
- ****Example:**** Employee and their personal details.

****B. One-to-Many:****

- One record in Table A is related to many records in Table B.
- ****Example:**** Department (one) and Employees (many).

****C. Many-to-Many:****

- Many records in Table A are related to many records in Table B.
- ****Example:**** Students and Subjects (a student can take many subjects, a subject can have many students).

3. Advantages of Relating Tables

- ****Reduces Data Redundancy:**** Data is stored only once.
- ****Improves Data Integrity:**** Enforces consistency.
- ****Easy to Maintain:**** Changes in one table reflect in related tables.
- ****Efficient Data Retrieval:**** Use queries to get data from multiple tables.

4. Creating Relationships

1. Go to Tools → Relationships.
2. Select the tables to relate.
3. Drag the primary key field from one table to the foreign key field in another table.
4. In the dialog box, confirm the relationship.
5. Click Create.

5. Referential Integrity

****Definition:**** A rule that ensures relationships between tables are maintained.

****Enforcing Referential Integrity:****

- Prevents deletion of a record if it is referenced in another table.
- Updates to the primary key are automatically reflected in foreign key fields.

****Steps to Enable:****

- In the Relationship dialog box, select "Enforce Referential Integrity."

CHAPTER 11: QUERIES IN BASE

1. What is a Query?

****Definition:**** A question asked to the database to retrieve specific data.

****Functions:****

- Retrieve data based on conditions.
- Filter and sort data.
- Combine data from multiple tables.
- Perform calculations.

2. Creating a Query Using Wizard

1. In the Database Pane, select Queries.
2. In the Tasks Pane, click "Use Wizard to Create Query."
3. Select the fields you want to include.
4. Set sorting criteria.
5. Set search conditions.
6. Select the type of query.
7. Click Finish.

3. Creating a Query Using Design View

1. In the Database Pane, select Queries.
2. In the Tasks Pane, click "Create Query in Design View."
3. Select the tables for the query.
4. Drag the fields from the tables to the grid.
5. In the grid:

- **Field:** Name of the field.
- **Table:** The table it comes from.
- **Sort:** Ascending/Descending.
- **Visible:** Show/hide the field.
- **Criteria:** The condition to filter data.

6. Run the query (F5 or the Run Query button).

Example:

To get students from Class 10:

- **Field:** Class
- **Criteria:** =10

4. Editing a Query

- Right-click on the query in the Database Pane.
- Select Edit.
- Make changes in the Design View.
- Save the query.

5. Working with Numerical Data in Queries

Using Criteria:

- **Greater than:** >100
- **Less than:** <100
- **Between:** BETWEEN 100 AND 200
- **Equals:** =100

Using Functions:

- **Sum:** Total of a column.
- **Average:** Average of a column.
- **Count:** Number of records.
- **Min/Max:** Minimum/Maximum value.

Example:

To get the total salary of all employees:

- In the Design View, add the Salary field.
- In the Functions row, select "Sum."

CHAPTER 12: FORMS AND REPORTS

1. Forms

****Definition:**** A user-friendly interface for entering, editing, and viewing data.

****Advantages:****

- Easy data entry.
- Prevents data entry errors.
- Shows only relevant fields.
- Can include pictures and instructions.

2. Creating a Form Using Wizard

1. In the Database Pane, select Forms.
2. In the Tasks Pane, click "Use Wizard to Create Form."
3. Select the table/query for the form.
4. Select the fields to include.
5. Select the layout and style.
6. Set the page orientation and size.
7. Click Finish.

****Example:****

Create a form for the Employee table to enter employee details.

3. Form Controls

****Definition:**** Elements on a form that allow user interaction.

****Common Controls:****

- ****Text Box:**** Enter text or numbers.
- ****Label:**** Display text.
- ****Push Button:**** Perform an action (Save, Cancel).
- ****Check Box:**** Select yes/no options.
- ****List Box:**** Select from a list.
- ****Combo Box:**** Drop-down list.
- ****Date/Time Field:**** Enter dates and times.
- ****Image Control:**** Display images.

4. Creating a Form in Design View

1. In the Database Pane, select Forms.
2. In the Tasks Pane, click "Create Form in Design View."
3. Use the Form Controls toolbar to add controls.
4. Set properties for each control (Data source, labels).
5. Save the form.

5. Reports

****Definition:**** A formatted output of data designed for printing or viewing.

****Features:****

- Grouping and summarising data.
- Calculations (totals, averages).
- Professional formatting (headings, borders, logos).

6. Creating a Report Using Wizard

1. In the Database Pane, select Reports.
2. In the Tasks Pane, click "Use Wizard to Create Report."
3. Select the table/query for the report.
4. Select the fields to include.
5. Group records (optional).
6. Sort records.
7. Choose the layout and style.
8. Add headings, titles, and page numbers.
9. Click Finish.

7. Adding Titles and Headings

- In the report, click on the title or heading.
- Edit the text.

8. Adding Date and Time

1. Go to Insert → Date and Time.
2. Select the format.
3. Click OK.

UNIT 4: MAINTAIN HEALTHY, SAFE AND SECURE WORKING ENVIRONMENT

CHAPTER 13: HEALTH, SAFETY AND SECURITY AT WORKPLACE

1. Workplace Hazards

****Definition:**** Anything that can cause harm to workers.

****Types of Hazards:****

- Physical Hazards
- Electrical Hazards
- Fire Hazards
- Health Hazards
- Stress at Work

2. Physical Hazards

****Definition:**** Hazards that can cause physical injury.

****Examples:****

- Slippery floors (slips and falls).
- Untidy work areas (trips and falls).
- Heavy lifting (back injuries).
- Poor lighting (eye strain).
- Moving parts of machinery (cuts, crushing).

****Prevention:****

- Keep work areas clean and tidy.
- Wear safety shoes and protective clothing.

- Use proper lifting techniques.
- Ensure good lighting.
- Use machine guards.

3. Electrical Hazards

****Definition:**** Hazards caused by electricity.

****Examples:****

- Exposed wires.
- Overloaded sockets.
- Damaged plugs and cables.
- Water near electrical appliances.

****Prevention:****

- Do not touch electrical equipment with wet hands.
- Check cables and plugs regularly.
- Do not overload sockets.
- Report damaged equipment immediately.
- Use surge protectors.
- Turn off equipment when not in use.

4. Fire Hazards

****Definition:**** Hazards that can cause fires.

****Examples:****

- Flammable materials (paper, cloth, chemicals).
- Faulty electrical wiring.
- Overheated equipment.
- Smoking in prohibited areas.

****Prevention:****

- Keep flammable materials away from heat sources.
- Store chemicals properly.
- Check electrical wiring regularly.
- Follow fire safety rules.
- Use fire extinguishers.

5. Health Hazards

****Definition:**** Hazards that can cause illness or health problems.

****Examples:****

- Poor ventilation (respiratory issues).
- Prolonged sitting (back pain, posture problems).
- Stress (anxiety, depression).
- Noise (hearing loss).
- Dust and fumes (allergies, breathing problems).

****Prevention:****

- Ensure good ventilation.
- Take breaks (stretch, walk).
- Manage stress (exercise, relaxation).
- Use earplugs for loud noise.
- Wear masks for dust and fumes.

6. Stress at Work

****Definition:**** Physical and mental strain caused by excessive workload or pressure.

****Causes:****

- Heavy workload and deadlines.
- Lack of support.
- Poor work-life balance.
- Conflicts with colleagues.
- Job insecurity.

****Effects:****

- Anxiety, depression.
- Fatigue, burnout.
- Headaches, sleeplessness.
- Low productivity, absenteeism.

****Prevention:****

- Manage time effectively.
- Prioritise tasks.
- Ask for help when needed.
- Take regular breaks.
- Maintain a healthy work-life balance.

- Talk to someone about problems.

7. Safety Guidelines

General Safety Rules:

- Be aware of your surroundings.
- Follow safety procedures.
- Report hazards and incidents.
- Use safety equipment.
- Keep work areas clean and tidy.
- Do not take shortcuts.
- Attend safety training.

CHAPTER 14: WORKPLACE QUALITY MEASURES

1. Air Quality

Factors:

- Dust, fumes, smoke.
- Poor ventilation.
- Chemical emissions.

Measures:

- Use air purifiers.
- Ensure proper ventilation.
- Avoid smoking indoors.
- Reduce use of chemicals.
- Use plants to improve air quality.

2. Water Quality

Factors:

- Contaminated water.
- Lead, chemicals, bacteria.

Measures:

- Use water purifiers.
- Store water in clean containers.
- Check water quality regularly.
- Avoid water wastage.

3. Office Ergonomics

****Definition:**** Designing the workspace to fit the worker.

****Ergonomic Tips:****

- ****Chair:**** Adjustable height, back support, armrests.
- ****Desk:**** Proper height, enough legroom.
- ****Computer Monitor:**** At eye level, arm's length away.
- ****Keyboard and Mouse:**** Positioned comfortably.
- ****Posture:**** Sit straight, feet flat, hands at 90 degrees.

4. Computer Health and Safety

****A. Eye Strain:****

- ****Causes:**** Prolonged screen time, poor lighting.
- ****Prevention:****
 - Use the 20-20-20 rule: Every 20 minutes, look at something 20 feet away for 20 seconds.
 - Adjust screen brightness and contrast.
 - Use anti-glare screens.
 - Blink frequently.
 - Get regular eye check-ups.

****B. Musculoskeletal Problems:****

- ****Causes:**** Poor posture, repetitive movements.
- ****Prevention:****
 - Maintain good posture.
 - Take breaks (stretch, walk).
 - Use ergonomic furniture.
 - Avoid repetitive movements.

****C. Repetitive Strain Injury (RSI):****

- ****Causes:**** Repeated movements (typing, mouse use).
- ****Prevention:****
 - Use ergonomic keyboard and mouse.
 - Take regular breaks.

- Stretch your hands and wrists.
- Avoid overexertion.

5. Workplace Cleanliness

Importance:

- Prevents accidents.
- Prevents illness (germs, bacteria).
- Improves morale and productivity.

Measures:

- Clean work areas daily.
- Dispose of waste properly.
- Use waste bins and segregate waste.
- Clean restrooms regularly.
- Keep kitchens clean.

CHAPTER 15: PREVENT ACCIDENTS AND EMERGENCIES

1. Types of Accidents

Common Accidents:

- Slips, trips, and falls.
- Cuts and bruises.
- Burns and scalds.
- Electric shocks.
- Fire-related injuries.
- Chemical spills.

Prevention:

- Follow safety rules.
- Keep work areas clean and tidy.
- Use safety equipment.
- Report hazards immediately.
- Do not rush.

2. Emergencies

****Definition:**** A sudden, unexpected situation that requires immediate action.

****Examples:****

- Fire.
- Natural disasters (earthquake, flood).
- Medical emergencies.
- Bomb threats.
- Chemical spills.

3. Fire Safety

****Fire Triangle:****

- ****Fuel:**** Anything that burns.
- ****Oxygen:**** Supports combustion.
- ****Heat:**** Ignites the fire.

****Fire Extinguisher Types:****

- ****Water:**** For paper and wood.
- ****CO₂:** For electrical fires.
- ****Dry Chemical:**** For flammable liquids and gases.

****Fire Safety Measures:****

- Install smoke detectors.
- Have fire extinguishers.
- Conduct fire drills.
- Know evacuation routes.
- Do not block fire exits.
- Do not overload sockets.

****What to do in a Fire:****

1. Raise the alarm.
2. Call emergency services.
3. Evacuate the building.
4. Use the nearest fire exit.
5. Do not use lifts.
6. If trapped, stay low and cover mouth.
7. Do not re-enter the building.

4. First Aid

****Definition:**** The immediate care given to a person who is injured or suddenly ill.

****First Aid Kit Contents:****

- Bandages, gauze, adhesive tape.
- Antiseptic wipes, ointment.
- Scissors, tweezers.
- Pain relievers.
- Burn cream.
- Gloves, masks.

****Common First Aid Procedures:****

****A. Cuts and Bleeding:****

1. Clean the wound with water.
2. Apply pressure with a clean cloth.
3. Apply antiseptic ointment.
4. Cover with a bandage.

****B. Burns:****

1. Cool the burn with cold water (not ice).
2. Do not apply creams or ointments.
3. Cover with a clean cloth.
4. Seek medical help.

****C. Electric Shock:****

1. Turn off power source.
2. Do not touch the person with bare hands.
3. Use a non-conductive object to separate them.
4. Check for breathing and pulse.
5. Perform CPR if needed.
6. Call for help.

****D. Breathing Difficulties (CPR):****

1. Check for responsiveness.
2. Call for help.
3. Open the airway (tilt head).
4. Check for breathing.
5. If not breathing, start chest compressions (30 compressions).
6. Give 2 rescue breaths.
7. Repeat until help arrives.

5. Evacuation Procedures

****Steps:****

1. Stop work immediately.
2. Turn off equipment.
3. Follow the evacuation route.
4. Use the nearest exit.
5. Do not use lifts.
6. Assemble at the designated meeting point.
7. Do not re-enter the building.

6. Electrical Safety

****Rules:****

- Do not use damaged equipment.
- Do not touch electrical equipment with wet hands.
- Do not overload sockets.
- Turn off equipment when not in use.
- Report damaged wiring immediately.

IMPORTANT PRACTICAL TASKS

LibreOffice Writer

1. ****Create and Apply Styles:**** Create a new paragraph style, character style, and page style. Apply them to a document.
2. ****Insert and Modify Images:**** Insert an image, resize it, crop it, and apply text wrapping.
3. ****Create a Table of Contents:**** Create headings, apply heading styles, and insert a Table of Contents.
4. ****Use Templates:**** Create a template and use it to create a new document.
5. ****Track Changes and Add Comments:**** Enable track changes, make edits, add comments, and review changes.
6. ****Compare Documents:**** Compare two versions of a document.

LibreOffice Calc

1. **Use Goal Seek:** Find the input value needed to achieve a target output.
2. **Create and Use Scenarios:** Create different scenarios and compare the results.
3. **Record a Macro:** Record a macro to format cells and run it.
4. **Link Spreadsheet Data:** Create references to other sheets and documents.
5. **Create Hyperlinks:** Create hyperlinks to other sheets and documents.
6. **Share and Review a Spreadsheet:** Enable sharing, record changes, add comments, and accept/reject changes.

LibreOffice Base

1. **Create a Table:** Create a table using the wizard and design view. Set a primary key.
2. **Enter and Edit Data:** Enter data into a table, edit it, and delete records.
3. **Create Relationships:** Create a one-to-many relationship between two tables.
4. **Create a Query:** Create a query using the wizard and design view. Add criteria to filter data.
5. **Create a Form:** Create a form using the wizard. Add form controls.
6. **Create a Report:** Create a report using the wizard. Add titles, headings, and date/time.

KEY DEFINITIONS

Styles: A set of formatting attributes applied to text, paragraphs, pages, etc.

Template: A pre-designed document with styles and formatting.

Table of Contents: A list of headings and page numbers at the beginning of a document.

Macro: A recorded sequence of actions to automate tasks.

Scenario: A set of input values used for what-if analysis.

Goal Seek: A tool that finds the input value for a desired output.

Database: An organised collection of related data.

DBMS: Software that manages databases.

Table: A structured collection of data with rows and columns.

Query: A question asked to the database to retrieve data.

****Form:**** A user-friendly interface for data entry.

****Report:**** A formatted output of data for printing.

****Primary Key:**** A unique identifier for each record in a table.

****Foreign Key:**** A field that refers to the primary key of another table.

****Referential Integrity:**** A rule that ensures relationships between tables are maintained.

****Hazard:**** Anything that can cause harm.

****Accident:**** An unplanned event that causes injury or damage.

****Emergency:**** A sudden situation requiring immediate action.

****First Aid:**** Immediate care given to an injured or ill person.

****Ergonomics:**** Designing the workspace to fit the worker.

****Green Jobs:**** Jobs that contribute to preserving the environment.

****Sustainable Development:**** Development that meets present needs without compromising future generations.

****3Rs:**** Reduce, Reuse, Recycle.
